

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EX-7101-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Reliability Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Describe the concept and indices of reliability.
b) A hydraulic system is comprised of five components having the following constant failure rates (times are in days):
 $\lambda_1 = 0.001$, $\lambda_2 = 0.005$, $\lambda_3 = 0.0007$, $\lambda_4 = 0.0025$ and $\lambda_5 = 0.00001$.
i) Determine the system MTTF and standard deviation
ii) Find the system design life if a 0.99 reliability.
2. A series system has 10 identical components. If the over all system reliability must be a least 0.99, what is the minimum reliability required of each component.

3. If $f(x) = \frac{C}{1+x^2}$, $-\infty < x < \infty$, then find C and show that its corresponding distribution function is

$$f(x) = \frac{1}{\pi} \tan^{-1} x + \frac{1}{2}$$

EX-7101-GS

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[2]

4.
 - a) Explain Markov analysis in detail. How reliability can be calculated from this model?
 - b) Derive an equation showing relation between reliability and MTTF.
5.
 - a) What is reliability testing? Discuss the types of tests used to evaluate a design.
 - b) Define and explain acceleration factor. How do you evaluate the acceleration factor?
6.
 - a) Discuss the effect of preventive maintenance of MTTF of a component.
 - b) Derive the relationship between hazard function and reliability function.
7. Show that if the hazard function is decreasing, the PDF, $f(t)$, is also a decreasing function and its mode must therefore occur at $t = 0$.
8. Write short notes on :
 - a) Parametric method
 - b) Markov process
